

Pseudo-compactness Numbers: Examples

Among the chain conditions considered in this monograph are these three: calibre, compact-calibre, and pseudo-compactness. For the first two of these we have presented, in Corollaries 4.7 and 8.6, respectively, quite satisfactory characterizations of those classes of infinite cardinals which can arise, for some (completely regular, Hausdorff) space X , as the class of infinite cardinals for which the chain condition in question fails for X .

We believe that an equally definitive result concerning classes of pseudo-compactness may be available, but we abandoned the search when we noticed that the general methods used for calibre and compact-calibre do not work here. The difficulty is not in finding, say for a given regular cardinal λ , a space which is pseudo- (α, α) -compact if and only if $\text{cf}(\alpha) \neq \lambda$ (cf. Theorem 8.1). The difficulty is in finding a space which avoids (as pseudo-compactness numbers) the elements of a given set of cardinals. In brief: pseudo-compactness properties are not preserved by finite products. In the present chapter we document this failure in various ways. There is, for example, Corollary 9.8: for $\alpha \geq \beta$ with $\alpha \geq \omega$ and $1 < m < \omega$ there is a completely regular, Hausdorff space X such that X^{m-1} is pseudo- (α, β) -compact and X^m is not pseudo- (α, β) -compact.

The examples depend heavily on elementary properties (discussed and derived in the Appendix) of the Stone–Čech compactification $\beta(\alpha)$ of the (discrete) space α and, more particularly, of the type $T(p)$ of p for $p \in \beta(\alpha)$.

We cite a natural question left unsettled (cf. Remarks 9.11). If $\alpha \geq \beta \geq \omega$ with β singular and $1 < m < \omega$, is there a set $\{X_k : k < m\}$ of completely regular, Hausdorff spaces such that $\prod_{k \in A} X_k$ is pseudo- (β, β) -compact for every non-empty $A \subset m$ with $A \neq m$, but $\prod_{k < m} X_k$ is not pseudo- (α, ω) -compact?

We see first that in Theorem 2.3(c') the condition $\pi w(X) < \alpha$ cannot be omitted.

9.1 Theorem. Let α be an (infinite) singular cardinal. There is a completely regular, Hausdorff space X such that X is pseudo- $(\text{cf}(\alpha), \text{cf}(\alpha))$ -compact, X is not pseudo- (α, α) -compact, and $\pi w(X) = \alpha$.

Proof. We set $X = N_{(\text{cf}(\alpha))^+}(\alpha)$. It is immediate from Theorem 8.1 that X has the required properties.

We begin our investigation into the product of pseudo- (α, β) -compact spaces with a simple result used in the proof of Lemma 9.5 below.

9.2 Lemma. Let $\alpha \geq \beta \geq \omega$, let X be a pseudo- (α, β) -compact space and K a compact space. Then $X \times K$ is pseudo- (α, β) -compact.

Proof. Let $\{U_\xi \times V_\xi : \xi < \alpha\} \subset \mathcal{T}^*(X \times K)$ and let $\bar{x}$ be an element of X such that

$$|\{\xi < \alpha : U \cap U_\xi \neq \emptyset\}| \geq \beta$$

for every neighborhood U of $\bar{x}$. If for every $y \in K$ there is a neighborhood $U(y) \times V(y)$ of $\langle \bar{x}, y \rangle$ such that

$$|\{\xi < \alpha : (U(y) \times V(y)) \cap (U_\xi \times V_\xi) \neq \emptyset\}| < \beta,$$

we choose $F \in \mathcal{P}_\omega(K)$ such that $K = \cup_{y \in F} V(y)$, we set $U = \cap_{y \in F} U(y)$, and we note that

$$|\{\xi < \alpha : U \cap U_\xi \neq \emptyset\}| < \beta.$$

Hence there is $\bar{y} \in K$ such that

$$|\{\xi < \alpha : (U \times V) \cap (U_\xi \times V_\xi) \neq \emptyset\}| \geq \beta$$

for every neighborhood $U \times V$ of $\langle \bar{x}, \bar{y} \rangle$.

The following lemma will be used in 9.4 (for β regular) and in 9.5 (for β singular).

9.3. Lemma. Let $\alpha \geq \beta \geq \omega$, $p \in \beta(\alpha)$ with $\|p\| = \beta$, $0 < m < \omega$,

$$\alpha \cup T(p) \subset X_k \subset \beta(\alpha) \quad \text{for } k < m,$$

and $X = \prod_{k < m} X_k$.

If $s : \beta \rightarrow X$ satisfies

$$|(\pi_k \circ s)[B]| = \beta \quad \text{for all } B \in [\beta]^\beta, k < m,$$

then there is $q \in X$ such that $|\{\eta < \beta : s(\eta) \in V\}| = \beta$ for every neighborhood V of q .

Proof. There is $A \in p \cap [\alpha]^\beta$; we write

$$A = \{\xi(\zeta) : \zeta < \beta\}$$

with $\xi(\zeta') \neq \xi(\zeta)$ for $\zeta' < \zeta < \beta$.

By induction over $\{k : 0 \leq k \leq m-1\}$ we define a set $\{B_k : k < m\}$ of sets such that

- (i) $B_k \in [\beta]^\beta$ for $k < m$,
- (ii) $B_{k+1} \subset B_k$ for $k < m-1$, and
- (iii) $\pi_k \circ s|_{B_k}$ is one-to-one for $k < m$.

We set $B = B_{m-1}$ and we write

$$B = \{\eta(\zeta) : \zeta < \beta\}$$

with $\eta(\zeta') \neq \eta(\zeta)$ for $\zeta' < \zeta < \beta$.

For $k < m$ we define $f_k : \alpha \rightarrow \alpha \subset X_k$ by

$$\begin{aligned} f_k(\xi(\zeta)) &= \pi_k \circ s(\eta(\zeta)) \quad \text{for } \xi(\zeta) \in A, \\ f_k(\xi) &= 0 \quad \text{for } \xi \in \alpha \setminus A, \end{aligned}$$

we set $q_k = \bar{f}_k(p)$, we note (from Lemma B.7 of the Appendix) that $q_k \in T(p) \subset X_k$, and we set

$$q = \langle q_k : k < m \rangle \in X.$$

Let $V = \prod_{k < m} V_k$ be a basic neighborhood of q and set

$$U = \cap_{k < m} \bar{f}_k^{-1}(V_k) \text{ and}$$

$$C = \{\zeta < \beta : \xi(\zeta) \in U\}.$$

For $\zeta \in C$ and $k < m$ we have

$$(\pi_k \circ s)(\eta(\zeta)) = f_k(\xi(\zeta)) = \bar{f}_k(\xi(\zeta)) \in V_k$$

and hence $s(\eta(\zeta)) \in V$. Since $\|p\| = \beta$ we have $|C| = \beta$, and for $\zeta', \zeta \in C$ with $\zeta' \neq \zeta$ we have $\eta(\zeta') \neq \eta(\zeta)$ and hence

$$|\{\eta < \beta : s(\eta) \in V\}| = \beta,$$

as required.

9.4 Lemma. Let $\alpha \geq \beta \geq \omega$ with β regular, $p \in \beta(\alpha)$ with $\|p\| = \beta$,

$$0 < m < \omega,$$

$$\alpha \cup T(p) \subset X_k \subset \beta(\alpha) \text{ for } k < m,$$

and $X = \prod_{k < m} X_k$. Then X is pseudo- (β, β) -compact.

Proof. We proceed by induction. In each case we let $\{U^\eta : \eta < \beta\}$ be a set of non-empty basic subsets of X , we let $s(\eta) \in U^\eta \cap \alpha^m$ for $\eta < \beta$, and we show there is $q \in X$ such that

$$|\{\eta < \beta : s(\eta) \in V\}| = \beta$$

for every neighborhood V of q .

We suppose either that $m = 1$, or that $m = n + 1$ and the statement holds for $m = n$. We consider two cases.

Case 1. There are $\bar{k} < m$, $\bar{B} \in [\beta]^\beta$ such that $|(\pi_{\bar{k}} \circ s)[\bar{B}]| < \beta$. Since β is regular there is $\bar{\xi} < \alpha$ such that

$$|\{\eta \in \bar{B} : (\pi_{\bar{k}} \circ s)(\eta) = \bar{\xi}\}| = \beta.$$

If $m = 1$, then ($\bar{k} = 0$ and) the statement holds with $q = \bar{\xi}$. If $m = n + 1$, the statement follows from the inductive hypothesis and the fact that $\{q \in X : q_{\bar{k}} = \bar{\xi}\}$ is (a copy of) $\prod \{X_k : k < m, k \neq \bar{k}\}$.

Case 2. $|(\pi_{\bar{k}} \circ s)[B]| = \beta$ for all $B \in [\beta]^\beta$, $k < m$. The statement then follows from Lemma 9.3.

9.5 Lemma. Let $\alpha \geq \beta \geq \omega$ with β singular, $p \in \beta(\alpha)$ with $\|p\| = \beta$, $0 < m < \omega$,

$$N_{(\text{cf}(\beta))^+}(\alpha) \cup T(p) \subset X_k \subset \beta(\alpha) \text{ for } k < m,$$

and $X = \prod_{k < m} X_k$. Then X is pseudo- (β, β) -compact.

Proof. We proceed by induction. In each case we let $\{U^\eta : \eta < \beta\}$ be a set of non-empty basic subsets of X , we let $s(\eta) \in U^\eta \cap \alpha^m$ for $\eta < \beta$, and we show there is $q \in X$ such that

$$|\{\eta < \beta : s(\eta) \in V\}| = \beta$$

for every neighborhood V of q .

We suppose either that $m = 1$, or that $m = n + 1$ and the statement holds for $m = n$. We consider two cases.

Case 1. There are $\bar{k} < m$, $\bar{B} \in [\beta]^\beta$ such that $|(\pi_{\bar{k}} \circ s)[\bar{B}]| \leq \text{cf}(\beta)$. We set

$$A = (\pi_{\bar{k}} \circ s)[\bar{B}] \text{ and } K = \text{cl}_{\beta(\alpha)} A$$

and we note that K is a compact subspace of $X_{\bar{k}}$. If $m = 1$ then ($\bar{k} = 0$ and) $\{s(\eta) : \eta \in \beta\} \subset K$; it is then clear that there is $q \in K$ such that

$$|\{\eta \in \bar{B} : s(\eta) \in V\}| = \beta$$

for every neighborhood V of q . If $m = n + 1$, then

$$\{s(\eta) : \eta \in \bar{B}\} \subset (\prod \{X_k : k < m, k \neq \bar{k}\}) \times K$$

and the statement follows from the inductive hypothesis and Lemma 9.2.

Case 2. $|(\pi_k \circ s)[B]| > \text{cf}(\beta)$ for all $k < m, B \in [\beta]^\beta$. We claim that $|(\pi_k \circ s)[B]| = \beta$ for all $k < m, B \in [\beta]^\beta$. For such k and B set $C = (\pi_k \circ s)[B]$, suppose $|C| < \beta$, and set

$$B(\xi) = \{\eta \in B : (\pi_k \circ s)(\eta) = \xi\} \text{ for } \xi \in C.$$

Since $\cup_{\xi \in C} B(\xi) = B$ we have $\sup_{\xi \in C} |B(\xi)| = \beta$. Let $\{\beta_\sigma : \sigma < \text{cf}(\beta)\}$ be a set of cardinal numbers such that

$$\beta_\sigma < \beta \text{ for } \sigma < \text{cf}(\beta), \text{ and}$$

$$\sum_{\sigma < \text{cf}(\beta)} \beta_\sigma = \beta,$$

for $\sigma < \text{cf}(\beta)$ choose $\xi(\sigma) \in C$ such that $|B(\xi(\sigma))| \geq \beta_\sigma$, and set

$$\bar{B} = \cup_{\sigma < \text{cf}(\beta)} B(\xi(\sigma)).$$

Then $|\bar{B}| = \beta$ and $(\pi_k \circ s)[\bar{B}] = \{\xi(\sigma) : \sigma < \text{cf}(\beta)\}$, a contradiction. The claim is established. The required statement follows from Lemma 9.3.

The proof is complete.

9.6 Theorem. Let $\alpha \geq \beta \geq \omega$ and $1 < m < \omega$.

(a) If β is regular there is a set $\{X_k : k < m\}$ of completely regular, Hausdorff spaces such that

$\prod_{k \in A} X_k$ is pseudo- (β, β) -compact for $A \subset m, A \neq m$, and

$\prod_{k < m} X_k$ is not pseudo- (α, ω) -compact.

(b) If β is singular, there is a set $\{X_k : k < m\}$ of completely regular, Hausdorff spaces such that

$\prod_{k \in A} X_k$ is pseudo- (β, β) -compact for $A \subset m, A \neq m$, and

$\prod_{k < m} X_k$ is not pseudo- $(\alpha, (\text{cf}(\beta))^+)$ -compact.

Proof. From Corollary B.11 there is $\{p_n : n < m\} \subset \beta(\alpha)$ such that

$$\begin{aligned}\|p_n\| &= \beta \quad \text{for } n < m, \text{ and} \\ T(p_{n'}) \cap T(p_n) &= \emptyset \quad \text{for } n' < n < m.\end{aligned}$$

In (a) we set

$$Y_n = \alpha \cup T(p_n) \quad \text{for } n < m,$$

and in (b) we set

$$Y_n = N_{(\text{cf}(\beta))^+}(\alpha) \cup T(p_n) \quad \text{for } n < m,$$

and (using these definitions) we set

$$X_k = \cup \{Y_n : n < m, n \neq k\} \quad \text{for } k < m.$$

Now let $A \in \mathcal{P}_m^*(m)$. There is $n \in m \setminus A$. In (a) we have

$$\alpha \cup T(p_n) \subset X_k \subset \beta(\alpha) \quad \text{for } k \in A,$$

and hence $\prod_{k \in A} X_k$ is pseudo- (β, β) -compact (by Lemma 9.4); and in (b) we have

$$N_{(\text{cf}(\beta))^+}(\alpha) \cup T(p_k) \subset X_k \subset \beta(\alpha) \quad \text{for } k \in A,$$

and hence $\prod_{k \in A} X_k$ is pseudo- (β, β) -compact (by Lemma 9.5).

Now for $\xi < \alpha$ we define $x(\xi) \in \prod_{k < m} X_k$ by

$$x(\xi)_k = \xi \quad \text{for } k < m,$$

we set $U^\xi = \{x(\xi)\}$ for $\xi < \alpha$, and we note that $\{U^\xi : \xi < \alpha\}$ is an (indexed) family of non-empty open subsets of $\prod_{k < m} X_k$.

In (a) we have $\cap_{k < m} X_k = \alpha$ and hence for $x \in \prod_{k < m} X_k$ there is a neighborhood V of x such that

$$|\{\xi < \alpha : V \cap U^\xi \neq \emptyset\}| < \omega;$$

it follows that $\prod_{k < m} X_k$ is not pseudo- (α, ω) -compact.

In (b) we have $\cap_{k < m} X_k = N_{(\text{cf}(\beta))^+}(\alpha)$, and hence for $x \in \prod_{k < m} X_k$ there is a neighborhood V of x such that

$$|\{\xi < \alpha : V \cap U^\xi \neq \emptyset\}| \leq \text{cf}(\beta);$$

it follows that $\prod_{k < m} X_k$ is not pseudo- $(\alpha, (\text{cf}(\beta))^+)$ -compact.

The proof is complete.

9.7 Theorem. Let $\alpha \geq \beta \geq \omega$ and $1 < m < \omega$.

(a) If β is regular there is a completely regular, Hausdorff space X such that

X^{m-1} is pseudo- (β, β) -compact, and

X^m is not pseudo- (α, ω) -compact.

(b) If β is singular there is a completely regular, Hausdorff space X such that

X^{m-1} is pseudo- (β, β) -compact, and

X^m is not pseudo- $(\alpha, (\text{cf}(\beta))^+)$ -compact.

Proof. To prove (a) we define Y_n and X_k as in (the proof of) Theorem 9.6(a), and to prove (b) we define Y_n and X_k as in (the proof of) Theorem 9.6(b); and using this definition we let X be the set

$$X = \bigcup_{k < m} (X_k \times \{k\})$$

with the disjoint union topology. (The singleton spaces $\{k\}$ are introduced to ensure that

$$(X_{k'} \times \{k'\}) \cap (X_k \times \{k\}) = \emptyset \text{ for } k' < k < m.)$$

The space X^{m-1} is the union of $m^{m-1} < \omega$ pairwise disjoint, open-and-closed subspaces S of the form

$$S = \prod_{k < m} (X_k \times \{k\})^{d_k},$$

with $d_k \geq 0$ for $k < m$, and $\sum_{k < m} d_k = m - 1$, and for each such space S there is $n(S) < m$ such that $d_{n(S)} = 0$. In (a) we have

$$\alpha \cup T(p_{n(S)}) \subset X_k \subset \beta(\alpha) \text{ for } d_k \neq 0,$$

and hence S is pseudo- (β, β) -compact (by Lemma 9.4); and in (b) we have

$$N_{(\text{cf}(\beta))^+}(\alpha) \cup T(p_n) \subset X_k \subset \beta(\alpha) \text{ for } d_k \neq 0,$$

and hence S is pseudo- (β, β) -compact (by Lemma 9.5). Thus in each case X^{m-1} is the union of finitely many pairwise disjoint open-and-closed pseudo- (β, β) -compact subspaces. It is then clear that X^{m-1} is pseudo- (β, β) -compact.

The space X^m contains an open-and-closed subspace homeomorphic to $\prod_{k < m} X_k$. In (a) the space $\prod_{k < m} X_k$ is not pseudo- (α, ω) -compact (by Theorem 9.6(a)) and hence X^m is not pseudo- (α, ω) -compact (by Theorem 2.5(b)(iv)); and in (b) the space $\prod_{k < m} X_k$ is not pseudo- $(\alpha, (\text{cf}(\beta))^+)$ -compact (by Theorem 9.6(b)) and hence X^m is not pseudo- $(\alpha, (\text{cf}(\beta))^+)$ -compact (by Theorem 2.5(b)(iv)).

The proof is complete.

9.8 Corollary. Let $\alpha \geq \beta \geq 2$ with $\alpha \geq \omega$ and $1 < m < \omega$. There is a completely regular, Hausdorff space X such that

X^{m-1} is pseudo- (α, β) -compact, and

X^m is not pseudo- (α, β) -compact.

Proof. From Theorem 2.1 it follows that a regular, Hausdorff space is pseudo- (α, n) -compact for $n < \omega$ if and only if it is pseudo- (α, ω) -compact. Thus it is enough to consider the case $\beta \geq \omega$.

For regular $\beta \geq \omega$ the statement follows from Theorem 9.7(a), and for singular $\beta \geq \omega$ the statement follows from Theorem 9.7(b).

We saw in Theorem 3.16 that if α is a cardinal such that $\text{cf}(\alpha) > \omega$ and $\{X_i : i \in I\}$ is a set of spaces such that X_J is pseudo- (α, α) -compact for all $J \in \mathcal{P}_\omega^*(I)$, then X_I is pseudo- (α, α) -compact. We see now that the analogous statement fails for cardinals α such that $\text{cf}(\alpha) = \omega$. Theorem 9.9 supplements and complements Corollary 3.11(iii), which treats the product of pseudo- (α, β) -compact spaces with α singular and $\alpha > \beta$.

9.9 Theorem. There is a set $\{X_k : k < \omega\}$ of completely regular, Hausdorff spaces such that

$\prod_{k < m} X_k$ is pseudo- (α, α) -compact for $0 < m < \omega$ if $\text{cf}(\alpha) = \omega$, and $\prod_{k < \omega} X_k$ is not pseudo- (α, α) -compact if $\text{cf}(\alpha) = \omega$.

Proof. From Corollary B.11 of the Appendix there is $\{p_k : k < \omega\} \subset U(\omega)$ such that

$$T(P_{k'}) \cap T(p_k) = \emptyset \text{ for } k' < k < \omega.$$

We set

$$Y_n = \omega \cup T(p_n) \text{ for } n < \omega, \text{ and}$$

$$X_k = \omega \cup \{Y_n : n < \omega, n \neq k\} \text{ for } k < \omega.$$

For $k < m < \omega$ we have

$$Y_m = \omega \cup T(p_m) \subset X_k \subset \beta(\omega),$$

and hence $\prod_{k < m} X_k$ is pseudo- (ω, ω) -compact by Lemma 9.4.

We note further that $\pi_\omega(\prod_{k < m} X_k) = \omega$; hence from Theorem 2.3(c') it follows that if $\alpha > \omega$ and $\text{cf}(\alpha) = \omega$ then $\prod_{k < m} X_k$ is pseudo- (α, α) -compact.

Now set $X = \prod_{k < \omega} X_k$; we show that X is not pseudo- (ω, ω) -compact. We set

$$U_k = \{x \in X : x_n = k \text{ for } n \leq k\} \text{ for } k < \omega,$$

we note that $\{U_k : k < \omega\} \subset \mathcal{T}^*(X)$, and we show that the (indexed) family $\{U_k : k < \omega\}$ is locally finite in X .

If $x \in U_n$, then U_n is a neighborhood of x and

$$|\{k < \omega : U_n \cap U_k \neq \emptyset\}| = 1 < \omega.$$

If $x \notin \bigcup_{n < \omega} U_n$, there is $\bar{n} < \omega$ such that $x_{\bar{n}} \neq x_0$; indeed, otherwise there is $q \in \beta(\omega)$ such that $x_n = q$ for all $n < \omega$, and from $\bigcap_{n < \omega} X_n = \omega$ we have $q \in \omega$ and hence $x \in U_q$. Let V_0 and $V_{\bar{n}}$ be disjoint neighborhoods in $\beta(\omega)$ of x_0 and $x_{\bar{n}}$, respectively, and set

$$U = (\pi_0^{-1}(V_0) \cap \pi_{\bar{n}}^{-1}(V_{\bar{n}})) \setminus (\bigcup_{k < \bar{n}} U_k).$$

Then U is a neighborhood of x , and

$$U \cap (\bigcup_{k < \omega} U_k) = \emptyset.$$

The proof that X is not pseudo- (ω, ω) -compact is complete. It follows from Theorem 2.3(c) that if $\alpha > \omega$ and $\text{cf}(\alpha) = \omega$, then X is not pseudo- (α, α) -compact.

We note that the space $X = \prod_{k < \omega} X_k$ defined in (the proof of) Theorem 9.9 is pseudo- (α, ω) -compact for all $\alpha > \omega$. Indeed, for $J \in \mathcal{P}_\omega^*(\omega)$ the space X_J is pseudo- (ω, ω) -compact and hence pseudo- (ω^+, ω) -compact; thus X is pseudo- (ω^+, ω) -compact (by Corollary 3.6(c)) and hence pseudo- (α, ω) -compact for all $\alpha > \omega$.

We combine the essential features of the spaces X_k ($k < \omega$) into a single space.

9.10 Theorem. There is a completely regular, Hausdorff space X such that

X^m is pseudo- (α, α) -compact for $0 < m < \omega$ if $\text{cf}(\alpha) = \omega$, and

X^ω is not pseudo- (α, α) -compact if $\text{cf}(\alpha) = \omega$.

Proof. From Corollary B.11 of the Appendix there is $\{p_k : k < \omega\} \subset \beta(\omega)$ such that

$$T(p_{k'}) \cap T(p_k) = \emptyset \text{ for } k' < k < \omega.$$

We choose $\{A_k : k < \omega\}$ such that

$$\bigcup_{k < \omega} A_k = \omega,$$

$$|A_k| = \omega \text{ for } k < \omega, \text{ and}$$

$$A_{k'} \cap A_k = \emptyset \text{ for } k' < k < \omega,$$

and we set

$$B_k = A_k \cup \bigcup_{n \neq k} (\hat{A}_k \cap T(p_n)),$$

$$C = \beta(\omega) \setminus \bigcup_{k < \omega} \hat{A}_k, \text{ and}$$

$$X = C \cup \left(\bigcup_{k < \omega} B_k \right).$$

We show that X^m is pseudo- (ω, ω) -compact for $0 < m < \omega$. For $I \in \mathcal{P}^*(m)$ and $f \in \omega^I$ we set

$$V(f) = \prod_{i \in I} B_{f(i)}$$

and we claim that $V(f)$ is pseudo- (ω, ω) -compact. Indeed, there is $\bar{n} \in \omega \setminus f[I]$ and for $i \in I$ there is

$$p_{\bar{n}, i} \in \hat{A}_{f(i)} \cap T(p_{\bar{n}}),$$

and since

$$T_{A_{f(i)}}(p_{\bar{n}, i}) = \hat{A}_{f(i)} \cap T(p_{\bar{n}}, i) \text{ for } i \in I$$

by Corollary B.9 of the Appendix and $A_{f(i)} \cup T_{A_{f(i)}}(p_{\bar{n}, i})$ is homeomorphic to $\omega \cup T(p_{\bar{n}})$, the space $V(f)$ is pseudo- (ω, ω) -compact for $f \in \omega^I$ by Lemma 9.4.

Now let $\{U(n) : n < \omega\}$ be a set of basic subsets of $(\beta(\omega))^m$, let I be a subset of m maximal with respect to the property that there are $f \in \omega^I$ and $A \in [\omega]^\omega$ such that

$$U(n)_i \cap B_{f(i)} \neq \emptyset \text{ for } i \in I, n \in A,$$

and set $J = m \setminus I$. (Our notation in what follows indicates that $I, J \neq \emptyset$. We omit the (routine) modifications required in case $I = \emptyset$ or $J = \emptyset$.) Since $V(f)$ is pseudo- (ω, ω) -compact and $\pi^I[U(n)] \cap V(f) \neq \emptyset$ for $n \in A$, there is $x \in X^I$ such that

$$|\{n \in A : V_I \cap \pi_I[U(n)] \neq \emptyset\}| = \omega$$

for every neighborhood V_I of x . We note that there is $y \in (\beta(\omega))^J$ such that

$$|\{n \in A : V \cap U(n) \neq \emptyset\}| = \omega$$

for every neighborhood $V = V_I \times V_J$ of $\langle x, y \rangle$. Indeed, otherwise, since $(\beta(\omega))^J$ is compact, there are finite $F \subset (\beta(\omega))^J$ and for $y \in F$ a

neighborhood $V(y) = V(y)_I \times V(y)_J$ of $\langle x, y \rangle$ such that

$$(\beta(\omega))^J = \bigcup_{y \in F} U(y)_J, \text{ and} \\ |\{n \in A : V(y) \cap U(n) \neq \emptyset\}| < \omega;$$

we set $V = \bigcap_{y \in F} V(y)_I$, we have that V is a neighborhood of x , and we have

$$|\{n \in A : V \cap \pi_I[U(n)] \neq \emptyset\}| < \omega,$$

a contradiction.

We claim that $y_i \in X$ for $i \in J$. If $y_i \notin X$, then there is $k < \omega$ such that $y_i \in \hat{A}_k$, and since

$$|\{n \in A : V_i \cap U(n)_i \neq \emptyset\}| = \omega$$

for every neighborhood V_i of y_i and $B_k \cap \hat{A}_k$ is dense in $\hat{A}_k$, we have

$$|\{n \in A : U(n)_i \cap B_k \neq \emptyset\}| = \omega.$$

This contradicts the maximality of I .

We have shown that there is $\langle x, y \rangle \in X^I \times X^J = X^m$ such that

$$|\{n < \omega : (V_I \times V_J) \cap U(n) \neq \emptyset\}| = \omega$$

for every neighborhood $V = V_I \times V_J$ of $\langle x, y \rangle$ in $(\beta(\omega))^m$. The proof that X^m is pseudo- (ω, ω) -compact is complete. Since $\pi w(X^m) = \omega$, it follows from Theorem 2.3(c') that if $\alpha > \omega$ and $\text{cf}(\alpha) = \omega$, then X^m is pseudo- (α, α) -compact.

We show that X^ω is not pseudo- (α, α) -compact if $\text{cf}(\alpha) = \omega$. For $k < \omega$ the space B_k is an open-and-closed subspace of X ; hence there is a continuous function g_k from X onto B_k . The function g , defined on X^ω by the rule

$$(g(x))_k = g_k(x_k),$$

is a continuous function from X^ω onto $\prod_{k < \omega} B_k$. Since B_k is homeomorphic to the space X_k of (the proof of) Theorem 9.9 and $\prod_{k < \omega} X_k$ is not pseudo- (α, α) -compact, the space $\prod_{k < \omega} B_k$ is not pseudo- (α, α) -compact. It follows from Theorem 2.5(b)(i) that X^ω is not pseudo- (α, α) -compact.

The proof is complete.

9.11 Remarks. The theorems proved above leave certain natural questions unanswered.

(a) Let $\alpha \geq \beta \geq \omega$ with β singular and let $1 < m < \omega$. Is there a set $\{X_k : k < m\}$ of completely regular, Hausdorff spaces such that

$\prod_{k \in A} X_k$ is pseudo- (β, β) -compact for $A \in \mathcal{P}_m^*(m)$, and

$\prod_{k < m} X_k$ is not pseudo- (α, ω) -compact?

We note that for regular β the corresponding question is answered (in the affirmative) by Theorem 9.6(a).

Now let $\alpha \geq \beta \geq \omega$ with β singular, let φ be a function from β onto $\text{cf}(\beta)$ such that $|\varphi^{-1}(\{\sigma\})| < \beta$ for $\sigma < \text{cf}(\beta)$, let $p \in \beta(\alpha)$ with $\|p\| = \beta$ and set $q = \bar{\varphi}(p)$ and $X = \alpha \cup T(p) \cup T(q)$. It can then be shown, by an argument similar to that of Lemma 9.5 but more delicate, that X^m is pseudo- (β, β) -compact for $0 < m < \omega$. It then follows, choosing $\{p_k : k < m\} \subset \beta(\alpha)$ such that

$$\|p\| = \beta \text{ for } k < m, \text{ and}$$

$$T(p_{k'}) \cap T(p_k) = \emptyset \text{ for } k' < k < m,$$

and setting

$$q_k = \bar{\varphi}(p_k) \text{ for } k < m,$$

$$Y_n = \alpha \cup T(p_n) \cup T(q_n) \text{ for } n < m, \text{ and}$$

$$X_k = \cup \{Y_n : n < m, n \neq k\} \text{ for } k < n,$$

that

$\prod_{k \in A} X_k$ is pseudo- (β, β) -compact for $A \in \mathcal{P}_m^*(m)$, and

$\prod_{k < m} X_k$ is not pseudo- $(\alpha, (\text{cf}(\beta))^+)$ -compact.

The desired stronger conclusion, that $\prod_{k < m} X_k$ is not pseudo- (α, ω) -compact, seems unavailable because we are uncertain whether $\cap_{k < \omega} X_k = \alpha$; in particular we are uncertain whether $(\varphi$ may be chosen so that)

$$T(q_{k'}) \cap T(q_k) = \emptyset \text{ for } k' < k < m.$$

Thus the following question, of interest in itself, is relevant to (a) above.

(b) Let β be a singular cardinal, $p \in \beta(\beta)$ with $\|p\| = \beta$ and φ a function from β onto $\text{cf}(\beta)$ such that $|\varphi^{-1}(\{\sigma\})| < \beta$ for $\sigma < \text{cf}(\beta)$. Does it follow that $\bar{\varphi}[T(p)] = U(\text{cf}(\beta))$?

Notes for Chapter 9

The spaces we have denoted $T(p)$ (with $p \in \beta(\alpha)$) were introduced by Frolik (1967a) in connection with his proofs (see Frolik 1967a, 1968)

of the statement, given without appealing to the continuum hypothesis or to any other special set-theoretic assumption, that the space $\beta(\omega) \setminus \omega$ is not homogeneous.

The constructions of the present chapter, several of which appear in Comfort (1979, 1980), are for the most part simple extensions of the arguments used by Comfort (1967) and Frolík (1967b) to prove (independently) the case $\alpha = \omega$ of Theorems 9.6 and 9.9.